

Cheat Sheet For Panda

Cheat Sheet For PANDAS



What is Python?

Pandas is a *powerful* and *flexible* open-source data analysis and manipulation library for Python.

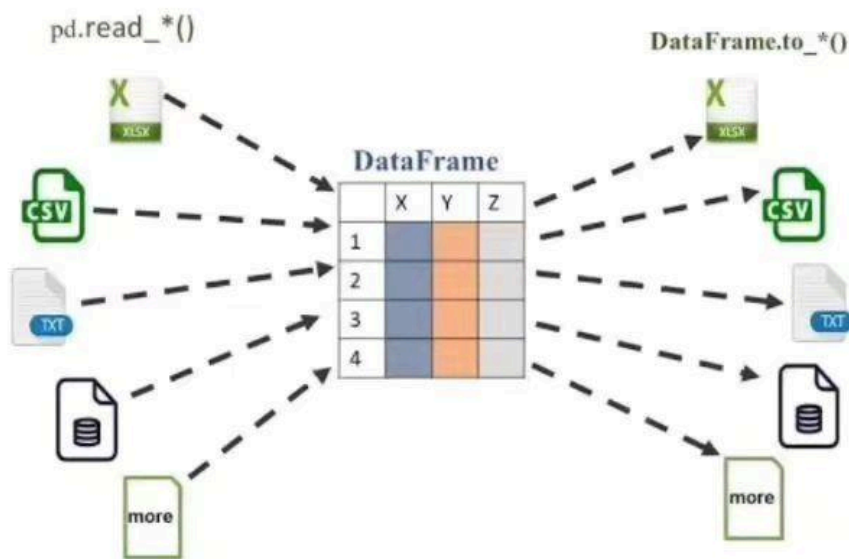
Important and Use-Cases?

Pandas is a popular Python library used in data science and analytics. It can handle large datasets and perform operations such as **cleaning**, **transformation**, and **exploration**. Applications include financial forecasting, customer segmentation, and machine learning data preprocessing."



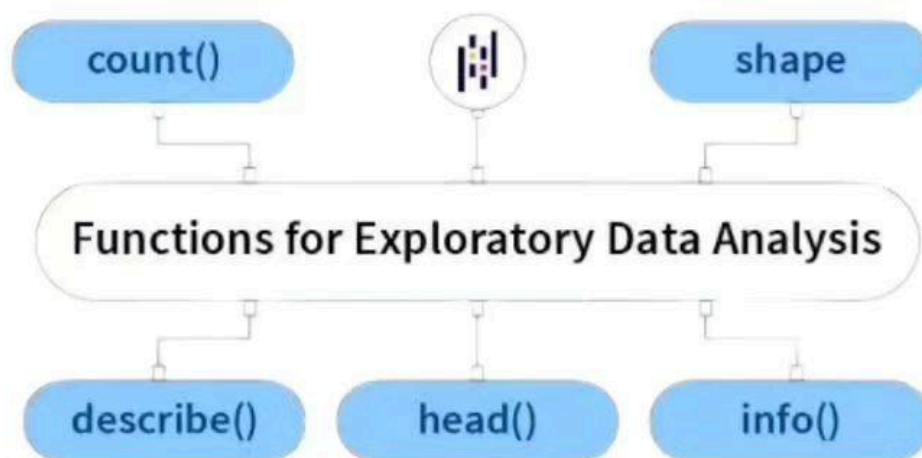
Reading & Writing Data

- **`pd.read_csv('file.csv')`**: Read a CSV file into DataFrame
- **`df.to_csv('file.csv')`** : Write a DataFrame to a CSV file
- **`pd.read_excel('file.xls')`** : Read an Excel file into a DataFrame
- **`df.to_excel('file.xlsx')`** : Write a DataFrame to an Excel file



Data Inspection

- **df.head():** Display the first 5 rows of a DataFrame
- **df.tail():** Display the last 5 rows of a DataFrame
- **df.info():** Display information about a DataFrame, including data types and memory usage
- **df.describe():** Display summary statistics of numerical columns in a DataFrame



Data Selection

- **df[col]:** Select a single column by name as a Series
- **df[[col1, col2]]:** Select multiple columns by name as a DataFrame
- **df.loc[row, col] :** Select a single value by row and column label
- **df.iloc[row, col]:** Select a single value by row and column index

Symbol	Industry	Shares
MSFT	Tech	100
GOOG	Tech	50
TSLA	Automotive	150



Industry = "Tech"
Shares < 100

GOOG	Tech	50
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Data Manipulation

- **df['new_col'] = value** : Add a new column to a DataFrame
- **df.drop (col, axis=1, inplace =True)** : Remove a column from a DataFrame
- **df.drop(row, axis=0, inplace=True)** : Remove a row from a DataFrame
- **df.sort_values(by=col, ascending=True)** : Sort a DataFrame by a column

	x1	x2	x3	x4	x5
0	x	15	a	20	foo
1	y	16	b	18	bar
2	x	17	c	16	foo
3	y	18	d	14	bar
4	x	19	e	12	foo
5	x	20	f	10	bar



Grouping

- **df.groupby(col).sum():** Group a DataFrame by a column and compute the sum of each group
- **df.groupby(col).median() :** Group a DataFrame by a column and compute the median of each group
- **df.groupby(col).max():** Group a DataFrame by a column and compute the maximum of each group
- **df.groupby(col).first():** Group a DataFrame by a column and return the first row of each group
- **df.groupby(col).size() :** Group a DataFrame by a column and return the size of each group

Team	Score
A	15
A	18
B	11
B	17
B	10
C	13



Team	Row Count
A	2
B	3
C	1

Python.hub

